

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

An artist paints and sells square tiles. The selling price P , in dollars, of a painted tile is a linear function of the side length of the tile s , in inches, as shown in the table below.

Side length, s (inches)	Price, P (dollars)
3	8.00
6	18.00
9	28.00

Which of the following could define the relationship between s and P ?

Answer

A. $P = 3s + 10$

B. $P = \frac{10}{3}s + 8$

C. $P = \frac{10}{3}s - 2$

D. $P = \frac{3}{10}s - \frac{1}{10}$

Correct Answer: C

Rationale

Choice C is correct. The relationship between s and P can be modeled by a linear equation of the form $P = ks + a$, where k and a are constants. The table shows that P increases by 10 when s increases by 3, so $k = \frac{10}{3}$. To solve for a , substitute one of the given pairs of values for s and P : when $s = 3$, $P = 8$, so $8 = \frac{10}{3}(3) + a$, which yields $a = -2$. The solution is therefore $P = \frac{10}{3}s - 2$.

Choice A is incorrect. When $s = 3$, $P = 8$, but $3(3) + 10 = 19 \neq 8$. Choice B is incorrect. This may result from using the first number given for P in the table as the constant term a in the linear equation $P = ks + a$, which is true only when $s = 0$. Choice D is incorrect and may result from using the reciprocal of the slope of the line.